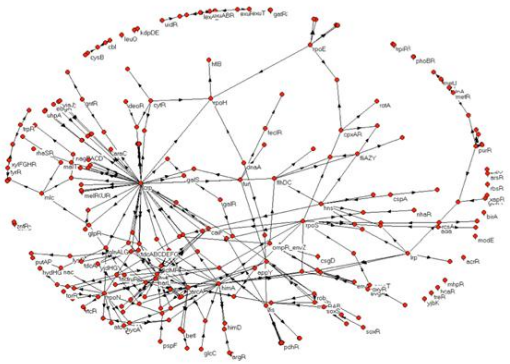
Computational Biology

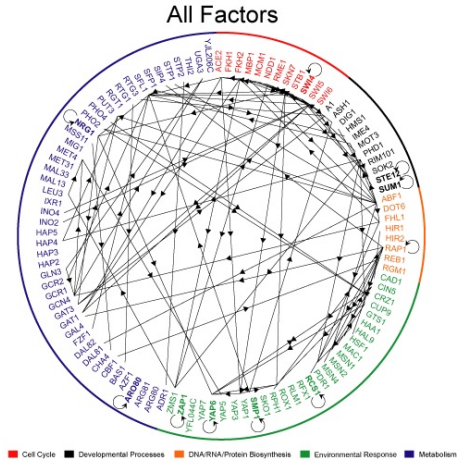
Lecture 17: *Gene Regulation Networks*



Evolution, DNA, Proteins

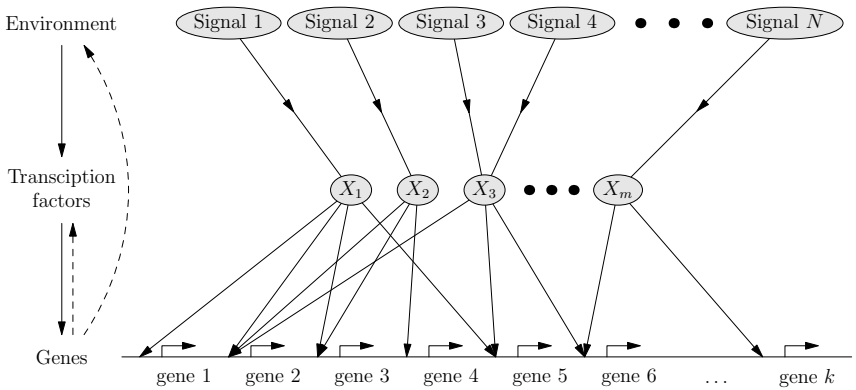
Outline

1. **Gene Regulation**
2. Chemical Equations
3. Motifs

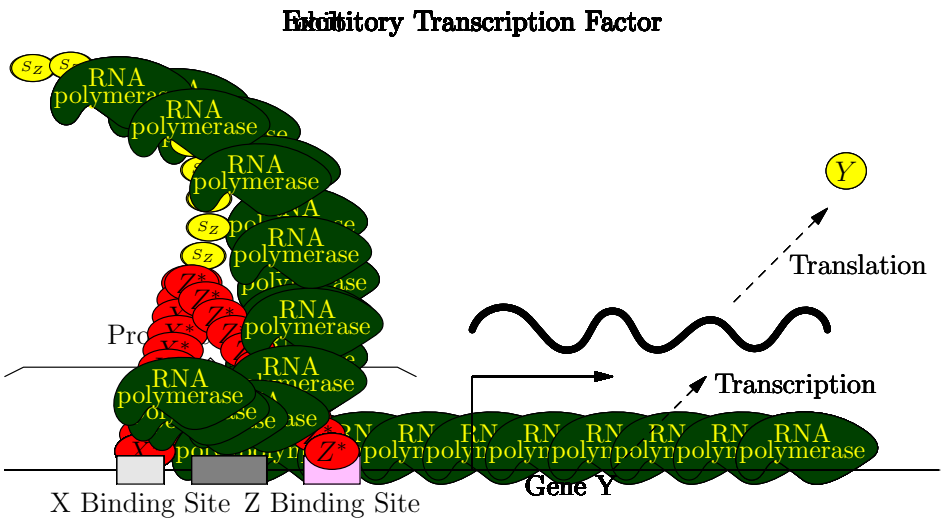


Transcription Networks

- Signalling molecules combine with transcription factors (proteins) which excite or inhibit the transcription of other proteins



Gene Transcription Regulation



Time Scales

- Reactions happen on different time scales
  - ⋆ Binding of small (signalling) molecules: ~ 1 msec
  - ⋆ Binding of active transcription factor: ~ 1 sec
  - ⋆ Transciption + translation of gene: ~ 5 min
  - ⋆ 50% change in concentration of protein: ~ 1 hour
- We can assume all events at earlier time scales happen instantly

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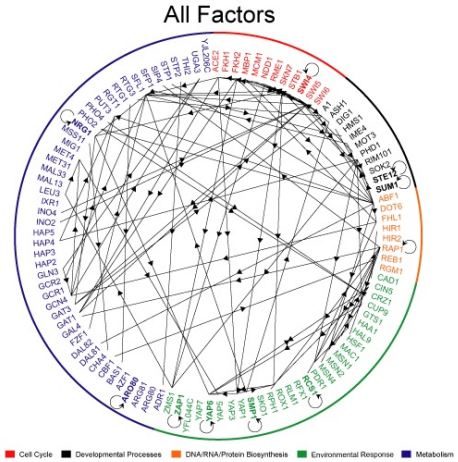


Chemical Equations

- We can write down the mechanism that underlie gene regulation in terms of chemical equations

$$X \xrightarrow{k_1} X + RNA_Y$$
$$RNA_Y \xrightarrow{k_2} RNA_Y + Y$$
$$RNA_Y \xrightarrow{k_3} \varnothing$$

- These are massively simplified (but we can ignore reactions that happen quickly)
- The rate of reaction depends on the number/concentration of the molecules on the left-hand side and the reaction rate k_i

Changes on Average

- Chemical equations happen at random, so the rate equations describe the average changes

$$X \xrightarrow{k_1} X + RNA_Y$$
$$RNA_Y \xrightarrow{k_2} RNA_Y + Y$$
$$RNA_Y \xrightarrow{k_3} \varnothing$$

- Let $[X]$ denote the (mean) concentration of X then

$$\frac{\mathrm{d}[RNA_Y]}{\mathrm{d}t} = k_1 [X] - k_3 [RNA_Y]$$
$$\frac{\mathrm{d}[Y]}{\mathrm{d}t} = k_2 [RNA_Y]$$

Reaction Rates and Concentrations

- When we consider reaction involving multiple molecular species, e.g. two proteins forming a complex

$$X + Y \xrightarrow{k_4} C$$

- The reaction rates depend on concentrations of X and Y (how likely is it for two molecules to collide)
- To find the concentration we have to divide by the volume of the cell, V
- To obtain concentrations in moles (we also have to divide by Avogadro's number A)

Changes on Average

- Thus

$$X + Y \xrightarrow{k_4} C$$

- Implies on average

$$\frac{\mathrm{d}[X]}{\mathrm{d}t} = \frac{\mathrm{d}[Y]}{\mathrm{d}t} = -k_4\,[X]\,[Y] \qquad \frac{\mathrm{d}X}{\mathrm{d}t} = \frac{\mathrm{d}Y}{\mathrm{d}t} = -\frac{k_4}{\Omega}\,X\,Y$$

$$\frac{\mathrm{d}[C]}{\mathrm{d}t} = k_4\,[X]\,[Y] \qquad \frac{\mathrm{d}C}{\mathrm{d}t} = \frac{k_4}{\Omega}\,X\,Y$$

- $\Omega = VA$, $[X] = X/\Omega$

Fluctuations

- In chemistry we are used to there being so many molecules that the average rate equations provide a very good approximation of what really happens
- This is often true in biology, but not always
- Cells are small and some proteins can occur in very small numbers
- We return to the problem of modelling systems with small number of molecules in the last lecture
- For the time being we take the differential equation model as an accurate description of our gene regulation network

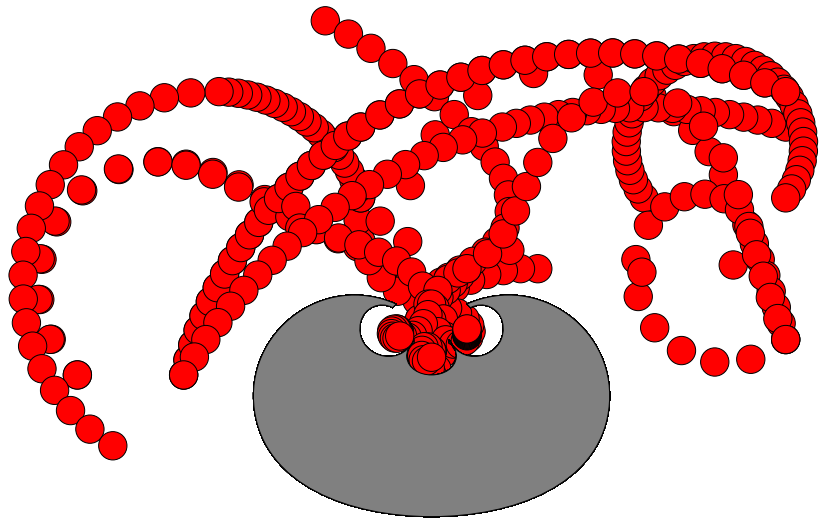
Switches

- Our molecular equation depend on the product of concentrations of the components on the left-hand side
- This can lead to strong non-linearities
- Such non-linearity can simulate switch like mechanism
- This is often useful for robust regulation

Signalling

- We showed earlier that proteins are often *activated* by signalling molecules combining with the protein to form a complex
- Proteins can have multiple sub-units that each bind with a signalling molecule
- A complex might only be stable when the protein has bound some number of signalling molecules

Multi-signalling



Hill Equation

- We assume that we form a complex with n signalling molecules S_X

$$n\,S_X + X \xrightarrow{k_{on}} C \qquad\qquad C \xrightarrow{k_{off}} n\,S_X + X$$

- Which we can model using

$$(K_X^n + [S_X]^n)\frac{d[C]}{dt} = k_{on}[X][S_X]^n - k_{off}[C] = k_{on}[X_T][S_X]^n - (k_{on}[S_X]^n + k_{off})[C] = 0$$

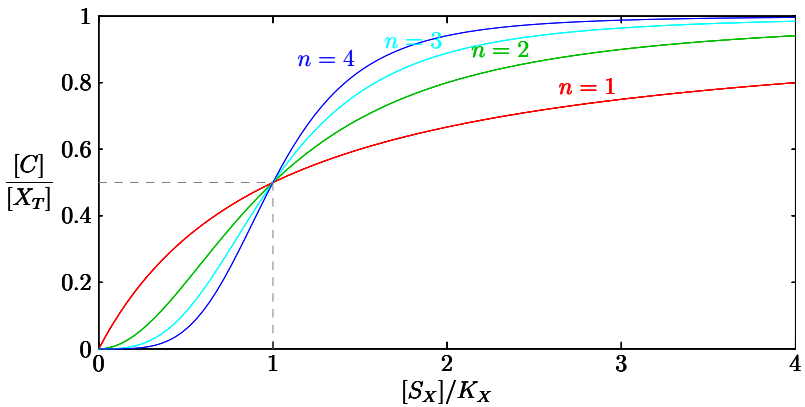
$$\frac{[C]}{[X_T]} = \frac{[S_X]^n}{K_X^n + [S_X]^n}$$

$$K_X = \left(\frac{k_{off}}{k_{on}}\right)^{1/n}$$

This equation is known as the Hill equation. The Hill coefficient n is the number of $[X]$ binding sites

Hill Coefficient

$$\frac{[C]}{[X_T]} = \frac{[S_X]^n}{K_X^n + [S_X]^n}$$



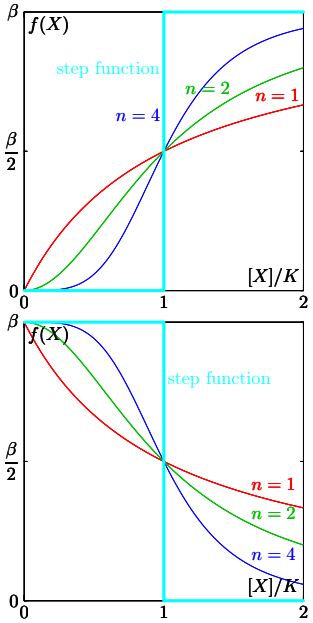
Activator and Repressor

- A protein that acts to increase the expression of a protein we call an **activator**
- The activation coefficient is often given by the Hill-equation

$$f([X]) = \frac{\beta\,[X]^n}{K^n + [X]^n}$$

- By a similar argument a protein that acts as a **repressor** is given by

$$f([X]) = \frac{\beta}{1 + ([X]/K)^n}$$



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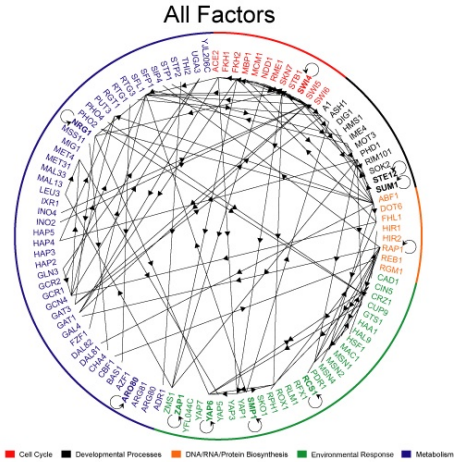
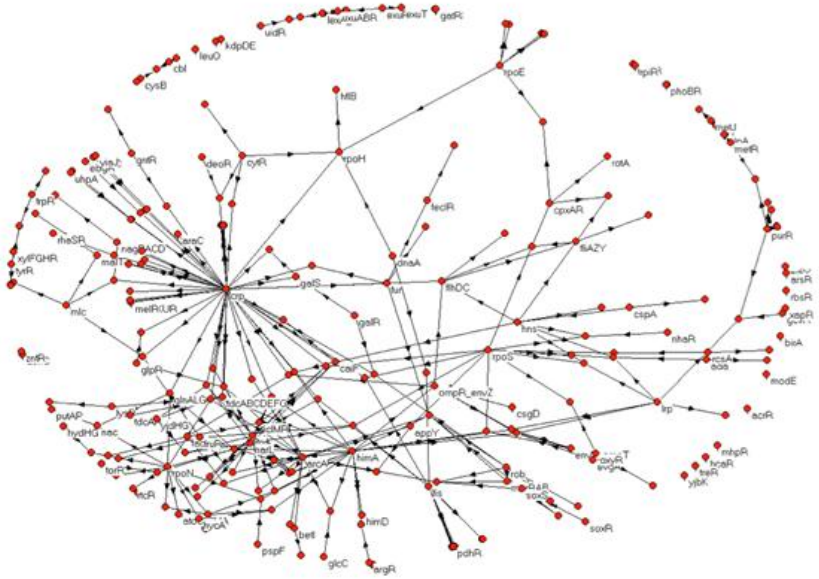


Understanding Gene Regulation Networks

- We can infer interactions between proteins as we described in a previous lecture
- Gene regulation networks tend to be rather complicated
- To make sense of them we wish to look for patterns, "**motifs**", that occur very often
- We look at interactions that occur more often than would be expected to happens in a random network

E-Coli GRN



Auto-regulation

- In E-coli we have $N = 424$ nodes and $E = 519$ edges
- The total number of directed edges (including self-edges) is N^2
- The number of self edges is N so the probability of a randomly chosen edge being a self-edge is $p = 1/N$
- The probability that if we choose E edges k will be self-edges is

$$\mathbb{P}\left(k|E,p\right) = \mathrm{Binom}(k|E,p) = \binom{E}{k} p^k \left(1-p\right)^{E-k}$$

- This has mean $\mathbb{E}\left[N_{\text{self}}\right] = E\,p \approx 1.2$ and standard deviation $\sigma = \sqrt{E\,p\,(1-p)} \approx 1.1$
- In fact we have $N_{\text{self}} = 40$ (that is, $32\,\sigma$ above the mean)

No Auto-Regulation

- To understand why auto-regulation is interesting let us first consider the case with no negative feedback
- We assume

$$\varnothing \xrightarrow{\hat{\beta}} \hat{X}$$
$$\hat{X} \xrightarrow{\hat{\alpha}} \varnothing$$

- So on average

$$\frac{\mathrm{d}[\hat{X}]}{\mathrm{d}t} = \hat{\beta} - \hat{\alpha}\,[\hat{X}]$$

- This has a steady state $[\hat{X}_{eq}] = \hat{\beta}/\hat{\alpha}$

Solving Differential Equations

- The full dynamics is given by the solution to the differential equation and depend on the initial condition, $[\hat{X}(0)]$
- Most differential equations cannot be solved in closed form
- However, they are usually quite easy to solve numerically (though it is difficult to understand how the solution depend on the parameters)
- Our equation is linear and is easy to solve. For $[\hat{X}(0)] = 0$

$$\frac{\mathrm{d}[\hat{X}(t)]}{\mathrm{d}t} = \hat{\beta} - \hat{\alpha}\,[\hat{X}(t)] \qquad \Rightarrow \qquad [\hat{X}(t)] = \frac{\hat{\beta}}{\hat{\alpha}}\left(1 - \mathrm{e}^{-\hat{\alpha}\,t}\right)$$

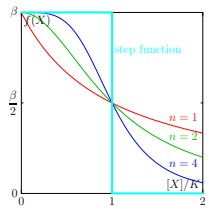
Auto-regulation

- We assume a negative feedback so that

$$\frac{\mathrm{d}[X]}{\mathrm{d}t} = f([X]) - \alpha\,[X]$$

- Where we have a Hill-type repressor

$$f([X]) = \frac{\beta}{1 + ([X]/K)^n} \approx \beta\;[\![[X] < K]\!]$$

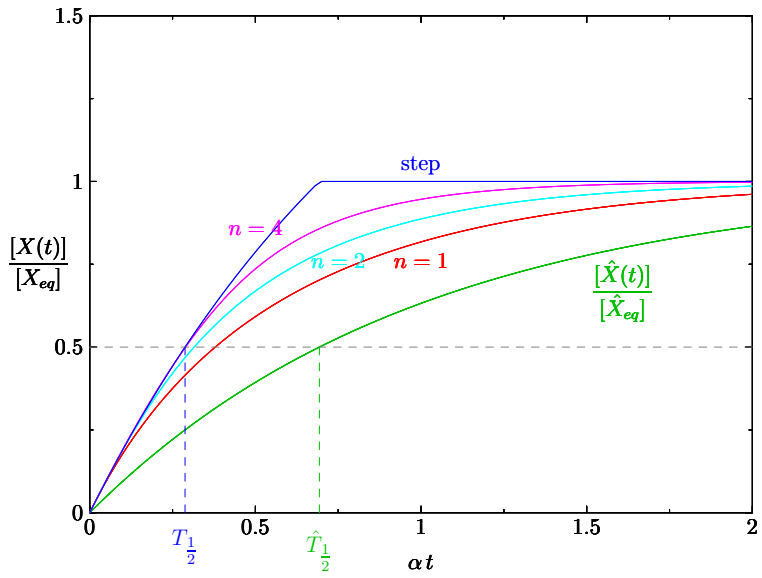


- Starting from $[X(0)] = 0$ initially $[X(t)] = \beta\,t$
- When $[X(t)] > K$ then $f([X(t)]) = f(K+\epsilon) \approx 0$ and a steady state is reached

Controlled Comparison

- To compare to having no feedback we want our two circuits to be as similar as possible
- Choose the same decay rate $\alpha = \hat{\alpha}$
- Choose $K = \hat{\beta}/\hat{\alpha}$ so that the steady state or equilibrium value of $X(t)$ is the same in both systems $[\hat{X}_{eq}] = [X_{eq}]$
- Note that in the auto-regulation system the rate of production β is decoupled from the steady-state

$\beta = 2$



Advantages of Auto-regulation

- Auto-regulation allows us to use a large production rate β for the same equilibrium, speeding up the equilibration time
- Note that we can independent evolve K (by mutations to the binding site of X in the promoter region) and β, by mutations in the binding site of RNA polymerase
- Another advantage is that β and α can strongly fluctuate during the life time of a cell caused by fluctuations in metabolic capacity
- K depends on the chemical bonds in the binding site and is typically more stable

Summary

- We can begin to understand some of the gene regulation dynamics by writing down chemical equations
- We can express the mean change in number of molecules through differential equations
- We can find common motifs, by looking for patterns that occur in GRN more often than we would expect in a random network
- One common motif is negative auto-regulation (negative feedback) that allows us to increase the response rate to reach the same equilibrium level of a molecule